# Guide: Mersenne prefix exits

Collatz Lean research project

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## What changed

A growing prefix previously prevented a fixed search depth from handling all Mersenne parameters. We now express the states after that prefix as

$$X - 7, \quad 3X - 10, \qquad X = 3^{4h+5}, \quad k = 6h + 4.$$

This permits searching the exponent modulo a power of two, instead of simulating a separately bounded prefix for each k.

## The first formal rule

For every $k = 1468 + 1536z$, the reduced inverse pair merges after $k + 12$ steps. The additional twelve steps are the same symbolic rule throughout the progression. The prefix length is unbounded.

This is useful for the transfer program: a smaller transfer instance at $v = 8w + 5$, where $w = (2^k - 7)/9$, can prove the instance at $u = 2^k - 1$. It is a conditional step, not a proof that every Collatz orbit reaches one.

## Where to inspect it

The formal source is `lean/Collatz/MersenneExit.lean`. The declarations `mersenne_exit_merge`, `mersenne_inverse_exit`, and `mersenne_inverse_exit_progression` separate the small merge, the growing prefix, and their composition. The declarations `mersenne_exit_smaller` and `AffineTransfer.mersenne_exit` verify strict decrease and the conditional induction step.

The experiment and exact saved census are `analysis/mersenne_exit_search.py` and `analysis/mersenne_exit_results.json`. The scope and derivation are recorded in `analysis/MERSENNE_EXIT_CHECK.md`.

## Reproduce

From the repository root, run

```
python3 analysis/mersenne_exit_search.py --depth 16
python3 -m unittest discover -s analysis -p 'test_mersenne*.py'
python3 analysis/audit_theorems.py --build
```

From `lean/`, run `lake build Collatz.MersenneExit`. Lean and Lake must use the repository's pinned toolchain.

## What remains

The depth-16 census covers 110 of 4096 eligible exponent residues. The unresolved classes are not counterexamples. A full proof still needs transfers or reductions that cover every remaining case and make the induction well founded. The finite census has not been imported as a Lean theorem.